Predicting Natural Disaster Economic Impact from County Demographics and Housing Conditions

Project Phase 2: Progress Report

CSPC 5071 — Data Management for Data Science

Professor: Yueting Chen

February 27, 2026

Group SAD – Sravya, Anushka, Duy

Git Repository: https://github.com/dcnguyen060899/su_cpsc5071_group_project_sad_sravya_anushka_duy

1 Problem Statement

Throughout history, natural disasters have caused billions of dollars in economic damage across the United States. Their impact, however, varies significantly across communities. Data-wise, Counties with different demographic profiles, income levels, and housing conditions experience vastly different outcomes from similar events. From an initial data perspective, our central research question is:

> *Can we predict the level of federal disaster assistance allocated to a county based on its pre-disaster demographic characteristics and housing cost conditions?*

Initial data exploration and EDA have refined this question in three important ways.

First, we noticed that FEMA's public dataset does not include dollar amounts for assistance allocations. Our outcome variable is therefore limited to disaster declaration counts and program activation flags rather than continuous dollar estimates. This shifts the prediction task from estimating aid amounts to classifying or counting declaration frequency per county.

Second, because we are studying pre-disaster county characteristics rather than tracking outcomes over time, this is a cross-sectional study. Each observation represents a county at a point in time. The demographic and housing snapshots (Census 2022, HUD FY2025) serve as stable proxies for persistent county characteristics, while FEMA declarations (2020–2024) serve as the multi-year outcome of interest.

Third, EDA revealed that the target variable (`total_disasters` per county) is heavily right-skewed, with a mean of 5.7, a median of 4, and a maximum of 149 declarations. This suggests that log-transformation will be necessary for linear models. Additionally, geographic position is not captured by demographics alone, which is a fundamental modeling limitation we will address through state-level controls in the next phase.

2 Data Preprocessing

Our preprocessing pipeline integrates three federal datasets joined by standardized 5-digit county FIPS codes, implemented across two notebooks: an initial data exploration notebook and a database pipeline notebook that loads cleaned data into MySQL.

To us, we believe this step was the most critical in our process. By examining the raw data from multiple sources, we were able to clearly define the underlying data quality issues and structural inconsistencies. This understanding guided our database design decisions, ensuring that constraints across sources were mutually compatible while preserving as much information as possible, ultimately supporting meaningful modeling and a well-grounded answer to our research question.

2.1 Data Sources

Three federal datasets were integrated using 5-digit FIPS codes as the common join key. The OpenFEMA Disaster Declarations v2 dataset covers fiscal years 2020–2024 and contains 16,298 declarations

spanning 3,277 counties. The Census ACS 5-Year Estimates for 2022 cover all U.S. counties, providing demographic records for 3,222 counties. The HUD Fair Market Rents for FY2025 cover all U.S. counties and originally contained 4,764 rows, which were reduced to 3,228 unique county-level records after aggregation.

2.2 Data Cleaning

FEMA Disaster Declarations. The FEMA REST API is paginated at 1,000 records per request. Note that the reason we use pagination loop in the first place was because FEMA REST API enforces a hard cap of 1000 records there a pagination loop allow us to iteratively process that repeatedly queries, advancing through successive "pages" of data until all records are retrieved. To put another way, a more precise technical language, a pagination loop was implemented that increments the `$skip` OData parameter by 1,000 per request until an empty response is returned, retrieving all 16,298 records. FIPS codes were constructed by concatenating `fipsStateCode` and `fipsCountyCode` after zero-padding (e.g., `'41' + '067' = '41067'`). Date fields returned as ISO 8601 strings were parsed by extracting only the `YYYY-MM-DD` portion. The 28 raw columns were reduced to 14 analytically relevant fields, dropping internal metadata such as `hash`, `lastRefresh`, and `femaDeclarationString`.

Census ACS Demographic Data. The Census API returns all 3,222 U.S. counties in a single request. All numeric columns are returned as strings and were cast using `pd.to_numeric(..., errors='coerce')`. The Census Bureau uses −666,666,666 to indicate suppressed or unavailable data. One county had this value in `median_household_income` and it was replaced with `NaN` before any downstream analysis. FIPS codes were constructed by concatenating zero-padded `state` and `county` fields. Census API variable codes (e.g., `B01001_001E`) were renamed to human-readable labels: `total_population`, `median_household_income`, `poverty_count`, `unemployment_count`, `labor_force_count`.

HUD Fair Market Rent Data. The raw HUD file contains 4,764 rows for approximately 3,200 counties. Investigation revealed that 1,605 rows represent town-level subdivisions within New England counties, where HUD publishes separate FMR values per town, resulting in 1,536 duplicate 5-digit FIPS codes. This duplication constitutes a primary key violation for county-level analysis and would cause a many-to-many join explosion if unresolved. Specifically, if FEMA has 2 disaster rows for a county and HUD has 5 town rows for the same FIPS, a naive join produces $2 \times 5 = 10$ rows, inflating the disaster count fivefold. The solution was population-weighted averaging: for each county FIPS, all sub-county FMR values were aggregated into a single county-level value where each town's contribution is proportional to its population share. The `metro` flag was set to 1 if any sub-area within the county is classified as metropolitan. This reduced 4,764 rows to 3,228 unique county-level records.

2.3 Data Integration

Database Schema. The cleaned data was loaded into a MySQL relational database (`disaster_impact_db`) implementing a star schema with four tables. The `county` dimension table uses `FIPS CHAR(5)` as the primary key and was constructed from the union of all FIPS codes across all three sources (3,292 unique codes), establishing an authoritative registry of valid geographic entities. The `disasters` fact table stores one row per declaration event with foreign key references to `county`, containing disaster type, declaration date, incident type, and four boolean program flags. The `demographics` dimension stores one row per county from Census ACS 2022, and the `housing_costs` dimension stores one row per county from HUD FY2025.

Two SQL views were created. `county_disaster_summary` aggregates the fact table to one row per county. `analysis_ready` performs a full `LEFT JOIN` across all four tables, applying `COALESCE` so counties with no federal declarations show 0 rather than `NULL`.

Coverage Gaps. The `LEFT JOIN` (a type of outer join) produces three categories of incomplete records

despite the careful design to preserve information. First, 70 counties present in the FEMA or HUD datasets have no matching FIPS in the Census ACS, resulting in null demographic fields (an additional county has only `median_household_income` suppressed, totaling 71 nulls in that column). Second, 64 counties present in the FEMA or Census datasets have no matching FIPS in the HUD file after aggregation, resulting in null housing cost fields. Third, 15 counties recorded zero FEMA declarations over 2020–2024, for which `total_disasters` is set to 0 via `COALESCE` rather than left as `NULL`; these counties retain complete data and are usable for modeling. Because approximately 54 counties are missing *both* Census and HUD data, the union of all incomplete rows is 80, not $70 + 64 = 134$. Listwise deletion (referring to .dropna() in Pandas library) therefore drops 80 rows (2.4%), leaving **3,212** complete-case counties for predictive modeling.

2.4 Feature Engineering

The following derived features were created to normalize raw counts by county population size:

- `poverty_rate` = (`poverty_count` / `total_population`) $\times$ 100. Mean 14.6%, SD 7.5%, maximum 65.6%, driven by Native American reservation counties which are structural economic outliers rather than data errors.
- `unemployment_rate` = (`unemployment_count` / `labor_force_count`) $\times$ 100. Mean 5.2%, maximum 32.0%.
- `log_population` = $\log(1 +$ `total_population`$)$. Applied to compress the extreme right skew; median county population is approximately 26,000 while Los Angeles County has nearly 10 million.
- `fmr_2` selected as the single housing cost representative. All five FMR columns are correlated above 0.94 with each other. The 2-bedroom Fair Market Rent is HUD's primary published affordability benchmark and the most widely cited FMR in housing research.

3 Preliminary Results of EDA

3.1 Statistical Analysis

Dataset Overview. The `analysis_ready` view produces a flat dataset of 3,292 counties $\times$ 17 columns. The target variable `total_disasters` has a mean of 5.69, a standard deviation of 6.11, and ranges from 0 to a maximum of 149 declarations, with a median of 4. Total population spans from 50 to 9,936,690 with a mean of 103,777 and median of 26,118, reflecting extreme right skew. Median household income ranges from \$22,094 to \$170,463 with a mean of \$62,327 and a relatively clean unimodal distribution centered near \$59,926. Poverty rate averages 14.6% (SD 7.5%), unemployment rate averages 5.2% (SD 2.9%), and the 2-bedroom Fair Market Rent (`fmr_2`) averages \$1,051 ranging from \$503 to \$3,783.

Missing Value Audit. After the outer join, the dataset contains 71 missing values in `median_household_income` (2.2%), 70 missing values in all Census columns (2.1%), and 64 missing values in all HUD columns (1.9%). Profiling of the 80 counties with any NULL value shows that they have a higher mean disaster count (10.8) compared to the 3,212 retained counties (mean 5.4), and they are evenly split between metro and non-metro classifications (50% each), compared to retained counties which skew non-metro (61.3%). This mild selection bias is noted as a model limitation.

Target Variable Distribution. The target variable `total_disasters` is heavily right-skewed. The distribution has a mean of 5.69, a median of 4, and a maximum of 149 declarations. Most counties cluster at low declaration counts while a small number accumulate very high totals. The 2020 count is an outlier year driven by COVID-19 Biological incident declarations (approximately 9,700 declarations in 2020 versus 1,000–2,400 per year in 2021–2024). A log transformation $\log(1 +$ `total_disasters`$)$ produces a more symmetric distribution suitable for linear modeling. Public Assistance (`pa_count`)

accounts for nearly all declarations ($r = 0.989$ with `total_disasters`), while Individual Assistance (`ia_count`) shows zero variance in this period and will be excluded as uninformative.

Correlation Analysis. Pearson correlations between features and `total_disasters` reveal a clear hierarchy of predictive strength. The near-perfect correlation of `pa_count` ($r = +0.989$) reflects a structural property of FEMA's declaration categorization rather than a meaningful predictor; it will be excluded from modeling to prevent data leakage. Among substantive predictors, `poverty_rate` shows the strongest positive association ($r = +0.218$), followed by `unemployment_rate` ($r = +0.173$) and `log_population` ($r = +0.072$). On the negative side, `median_household_income` ($r = -0.126$) and `fmr_2` ($r = -0.063$) both show weak inverse relationships. Notable inter-feature correlations include `median_household_income` and `poverty_rate` ($r = -0.74$) and all FMR columns with each other ($r > 0.94$), informing the feature selection described in Section 2.4.

3.2 Data Visualizations

Target Variable Distribution. Three visualizations were produced for `total_disasters`: a raw count histogram confirming the heavy right skew, a log-transformed histogram showing an approximately symmetric distribution, and a box plot ($Q1 = 3$, $Q3 = 7$, IQR $= 4$) revealing numerous high-count outliers. These confirm that log-transformation is appropriate for linear regression and that tree-based methods may handle raw counts without transformation.

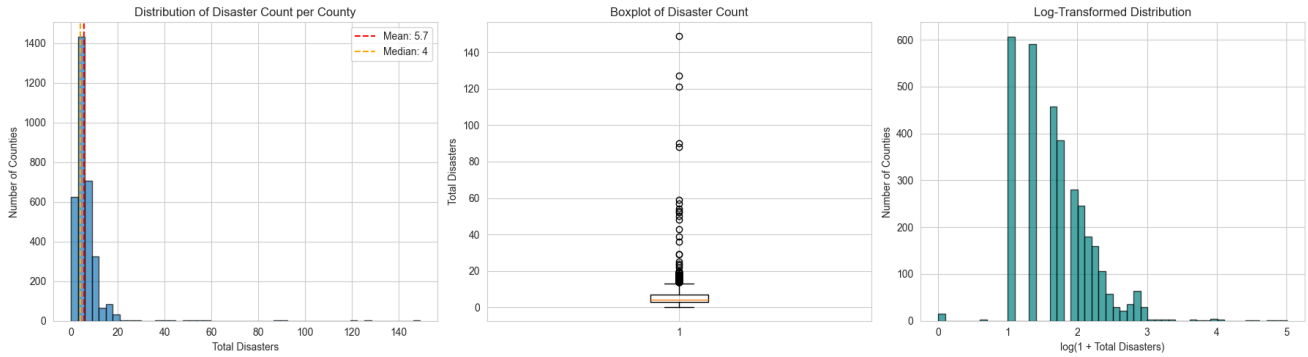

Figure 1: Distribution of `total_disasters`

Predictor Distributions. Histograms were produced for all demographic and housing predictors. Total population is extremely right-skewed and benefits from log transformation. Median household income is the cleanest predictor, approximately unimodal and centered near $60,000. Poverty rate and unemployment rate are moderately right-skewed, with extreme values at 50–65% and 32% respectively, driven by tribal reservation counties that are structural outliers. All FMR columns are right-skewed, with `fmr_2` ranging from $503 to $3,783.

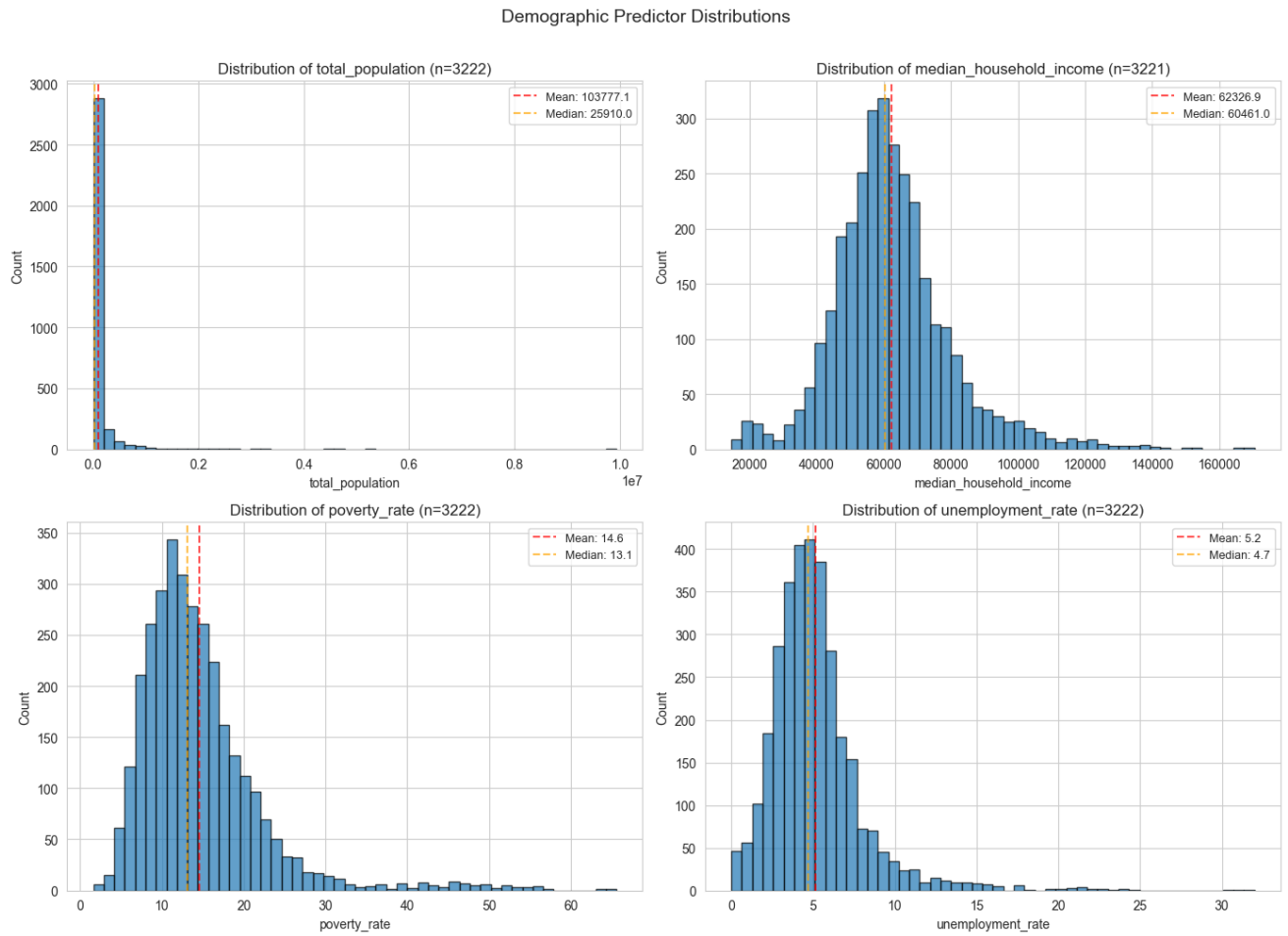

Figure 2: Distribution of demographic predictor variables across 3,222 U.S. counties.

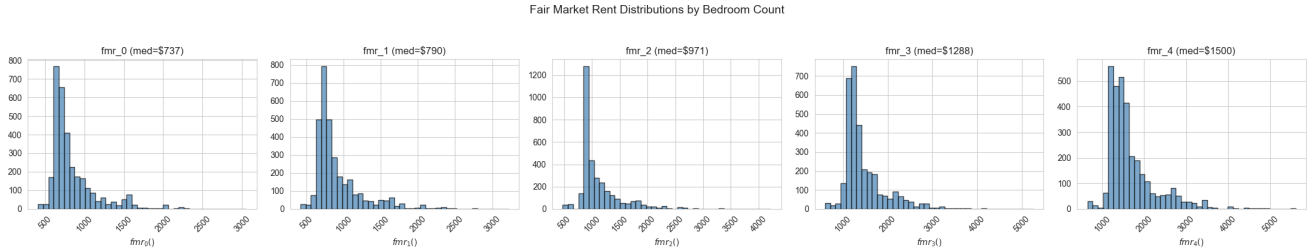

Figure 3: Fair Market Rent Distributions by Bedroom Count

Correlation Heatmap. A full correlation heatmap confirms the multicollinearity structure described above. The FMR columns form a tight cluster with pairwise correlations of 0.94–0.99. The demographic predictors show moderate inter-correlations (`poverty_rate` and `unemployment_rate`: $r = 0.65$). Disaster columns form a near-perfect cluster (`total_disasters` and `pa_count`: $r = 0.99$).

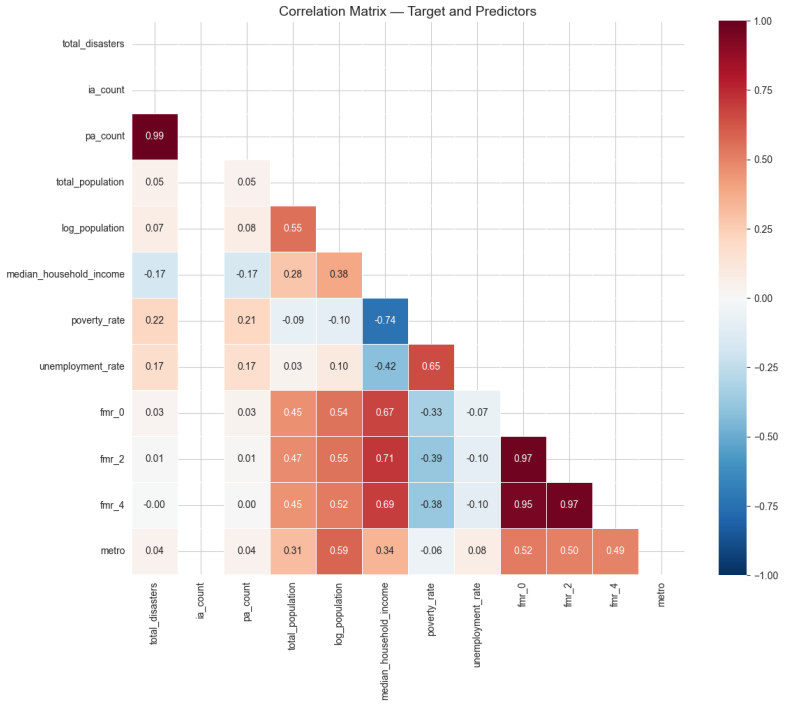

Figure 4: Correlation Matrix — Target and Predictors

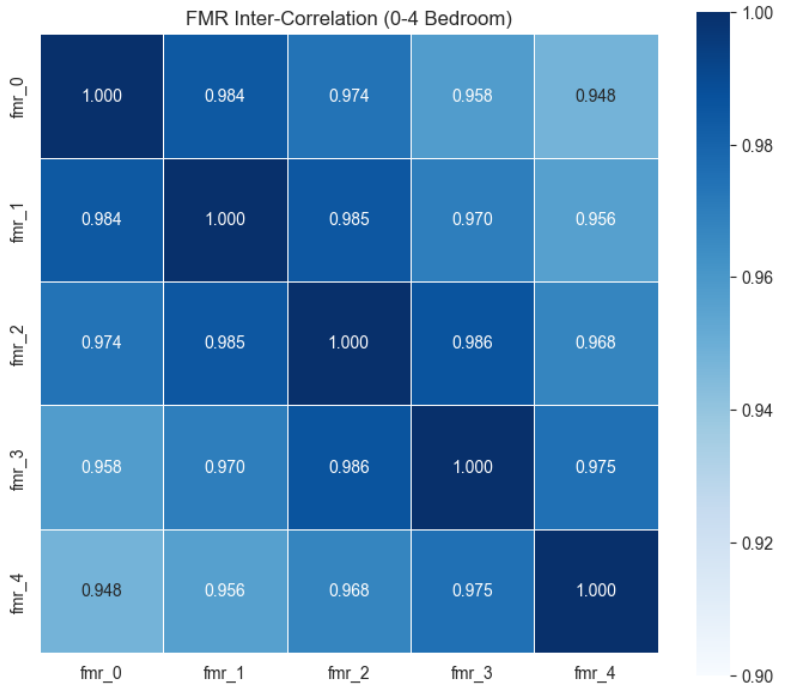

Figure 5: FMR Inter-Correlation (0–4 Bedroom)

Geographic Patterns. A state-level aggregation of mean disaster declarations per county shows that Maine leads (approximately 40 mean declarations per county), followed by Gulf Coast and tornado-alley states. This geographic concentration highlights the key limitation of our feature set: a county's location on the Gulf Coast hurricane track is not encoded in its poverty rate or median income. We will test the impact of this omission in the modeling phase by comparing models with and without state-level indicator variables.

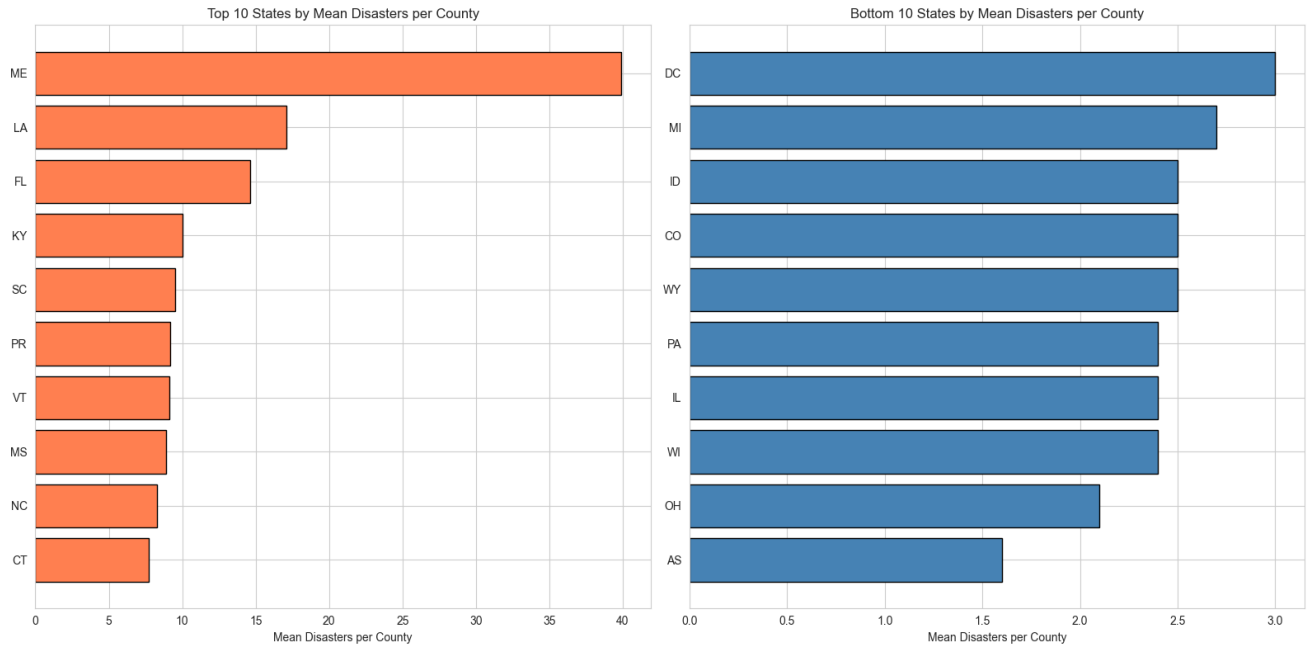

Figure 6: Top 10 states by mean disasters per county

Bivariate Scatter Plots. Scatter plots with linear trend lines were produced for each predictor against `total_disasters`. `log_population` shows a positive trend with wide scatter. `poverty_rate` shows the strongest positive trend among demographic predictors ($r = 0.218$). `median_household_income` shows a negative trend consistent with its inverse relationship with poverty. `fmr_2` shows a near-flat trend, suggesting housing cost alone is a weak predictor of disaster frequency.

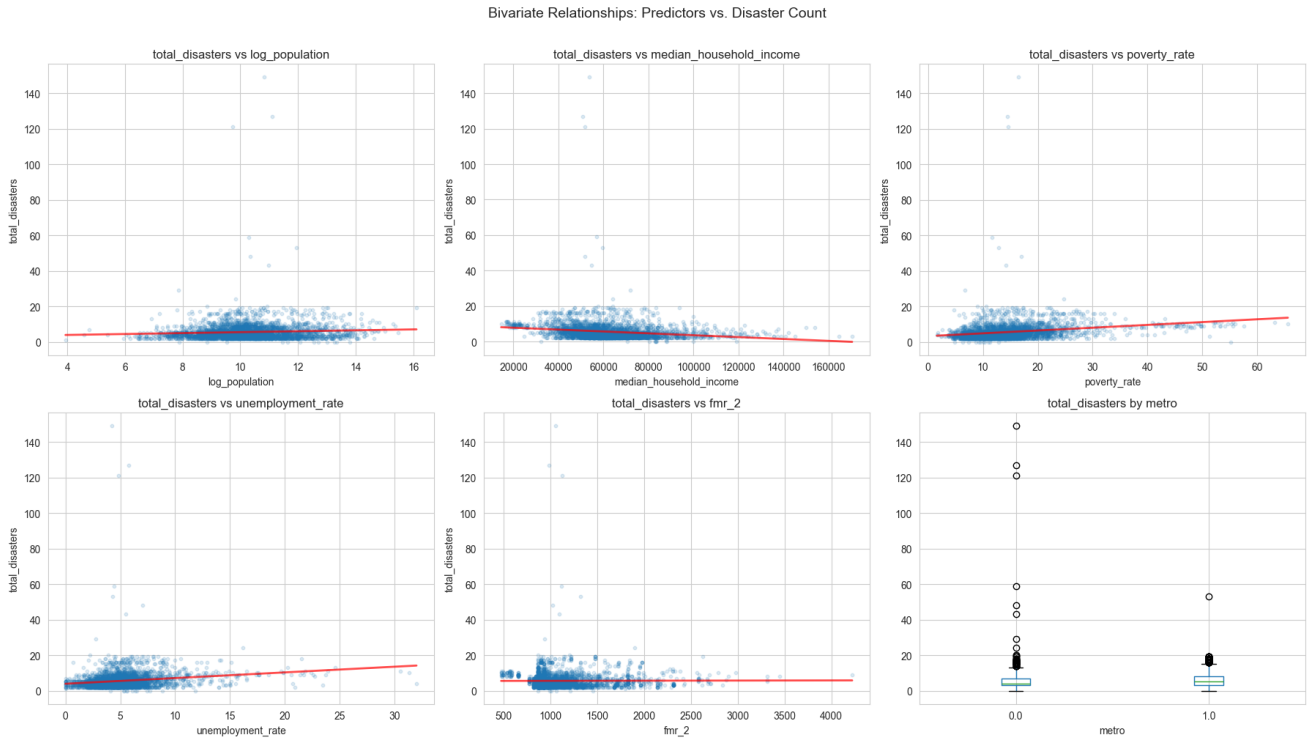

Figure 7: Bivariate Relationships: Predictors vs. Disaster Count

4 Plan for Completion

4.1 Remaining Data Processing Tasks

Remove the approximately 80 counties with missing Census or HUD data to produce the complete-case modeling dataset (approximately 3,212 rows). The finalized feature set consists of: `log_population`, `median_household_income`, `poverty_rate`, `unemployment_rate`, `fmr_2`, and `metro`. Conduct a COVID-19 sensitivity check by running analysis with and without the Biological incident type. Finalize the target variable: regression on $\log(1 + \texttt{total_disasters})$ versus binary classification at the median threshold of 4 declarations.

4.2 Analysis and Modeling Plan

1. Baseline models: OLS linear regression on the log-transformed target; logistic regression on the binary target.
2. Tree-based models: Decision tree for interpretability and non-linear relationships; feature importance ranking via Gini importance.
3. Geographic confounding test: compare model R^2 and feature importances with and without state-level dummy variables.

4. Class imbalance handling: apply SMOTE or class weighting if the binary target shows significant imbalance.

4.3 Evaluation and Validation Plan

An 80/20 stratified train/test split will be used with `random_state=42` for reproducibility. Regression performance will be evaluated using R^2 on the log-transformed target. Classification performance will be evaluated using accuracy, precision, recall, F1-score, and AUC-ROC. Residuals will be examined for geographic clustering to assess the impact of unmeasured location variables.

4.4 Documentation and Reporting Tasks

The final deliverable will be a comprehensive project report (Phase 3) structured as follows: an updated problem statement reflecting modeling results, a complete methodology section covering the full pipeline from data collection through model evaluation, a results section presenting model performance and key findings, and a discussion section addressing limitations and implications. All analysis code will be organized in the Git repository with three executable Jupyter notebooks: `initlal_data_exploratory.ipynb` (data exploration), `database_pipeline.ipynb` (ETL and database construction), and `eda.ipynb` (exploratory data analysis). Each notebook will include inline markdown cells explaining the reasoning behind each analysis step. The repository README will be updated to document the full project lifecycle and reproduction instructions.

4.5 Timeline for Completion

Week	**Date Range**	**Deliverables**
Week 9	Mar 2 – Mar 8	Finalize modeling notebook; baseline OLS and logistic regression; decision tree model; state fixed effects analysis; model comparison
Week 10	Mar 9 – Mar 15	Draft final report; results interpretation and limitations
Week 11	Mar 16 – Mar 19	Final report review, repository cleanup, and submission

How we plan to complete: Duy leads modeling implementation and report drafting; Sravya leads model evaluation and visualization; Anushka leads documentation review and repository organization. All members review the final report before submission.

References

[1] Cutter, S. L., Boruff, B. J., & Shirley, W. L. (2003). Social vulnerability to environmental hazards. *Social Science Quarterly*, 84(2), 242–261.

[2] Bakkensen, L. A., Fox-Lent, C., Read, L. K., & Linkov, I. (2017). Validating resilience and vulnerability indices in the context of natural disasters. *Risk Analysis*, 37(5), 982–1004.

[3] Yabe, T., & Ukkusuri, S. V. (2020). Effects of income inequality on disaster recovery: A case study of Hurricane Harvey. *Disasters*, 44(2), 256–286.

[4] Federal Emergency Management Agency. (2023). OpenFEMA Dataset: Disaster Declarations Summaries—v2. https://www.fema.gov/about/openfema/data-sets.

[5] Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, 321–357.

[6] U.S. Census Bureau. (2023). American Community Survey 5-Year Estimates (2019–2023). U.S. Department of Commerce. https://www.census.gov/programs-surveys/acs.

[7] Federal Register. (2022). Change to county-equivalents in the state of Connecticut, 87 FR 34235. https://www.federalregister.gov/documents/2022/06/06/2022-12063/.

[8] U.S. Department of Housing and Urban Development. (2023). Fair Market Rents. HUD USER. https://www.huduser.gov/portal/datasets/fmr.html.